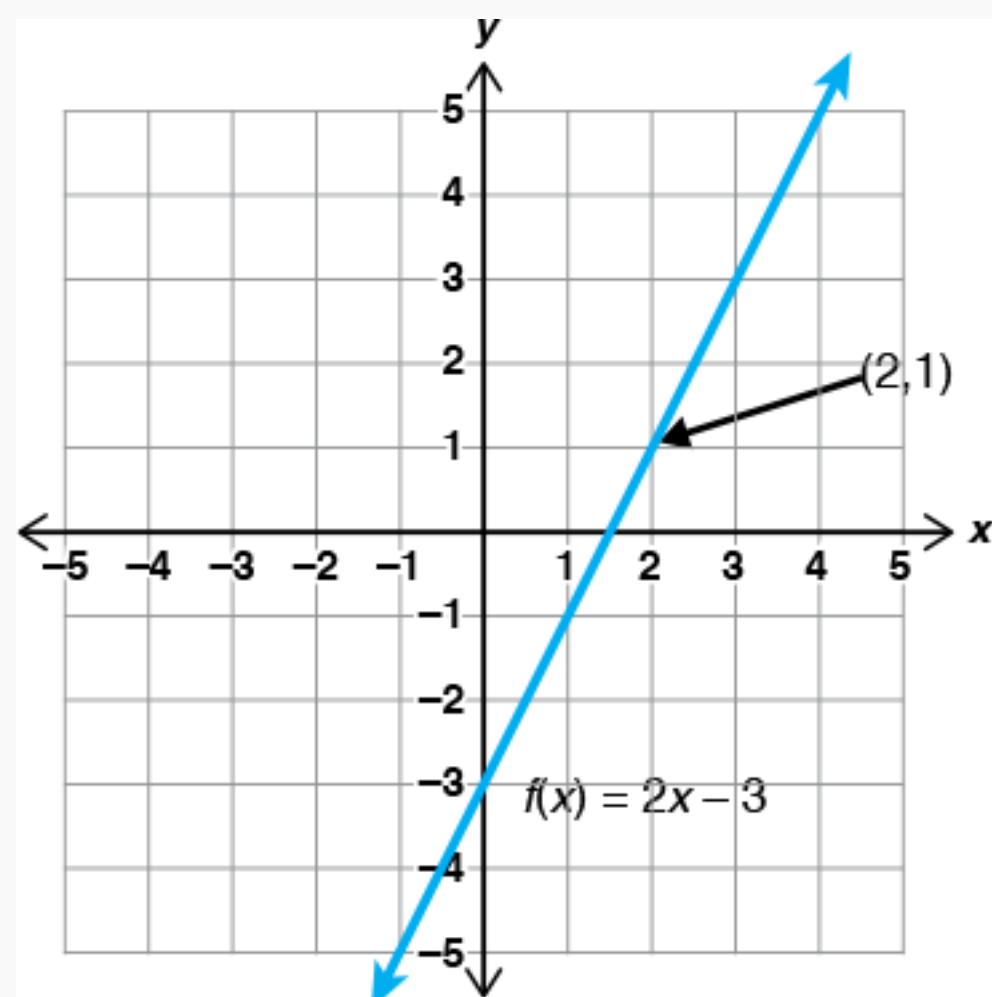


Understanding a Graph of a Function

Like an equation, a function can be graphed on a coordinate grid.

Example 1

For $f(x) = 2x - 3$, find $f(2)$.



Read the graph to find $f(2)$. What is the y -coordinate of the line when the x -coordinate is 2?

The y -coordinate is 1, so $f(2) = 1$.

The answer makes sense.

Try substituting 2 for x .

$$\begin{aligned} f(2) &= 2(2) - 3 \\ &= 4 - 3 \\ &= 1 \end{aligned}$$

The answer is the same.

A graph can also help you decide whether a relation is a function. Remember, if a relation is a function, every input can have only one output. That means that a graph of a function will have only one y -coordinate for every x -coordinate.

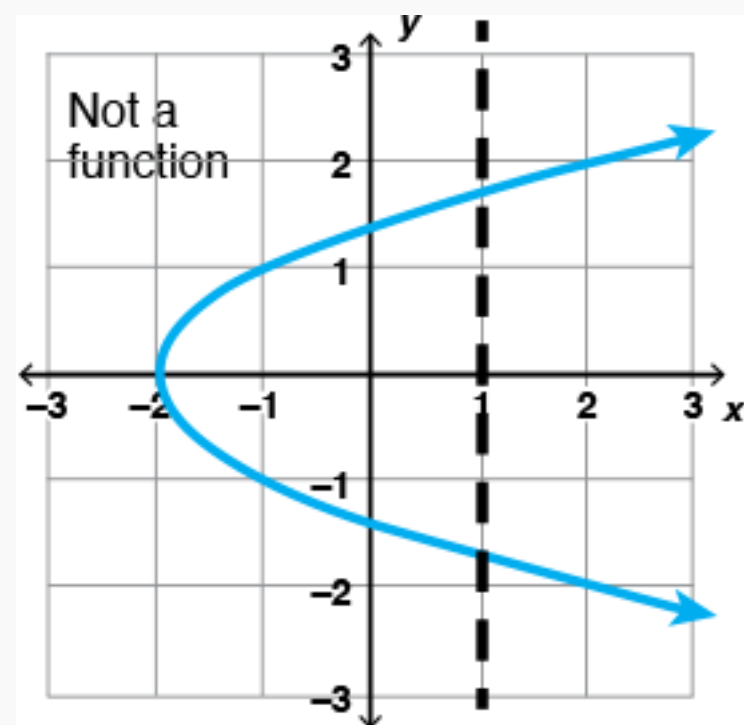
The **vertical line test** is a quick way to judge whether a graph represents a function. If you can draw a vertical line that passes through more than one point anywhere on a graph, the graph is not a function.

Example 2

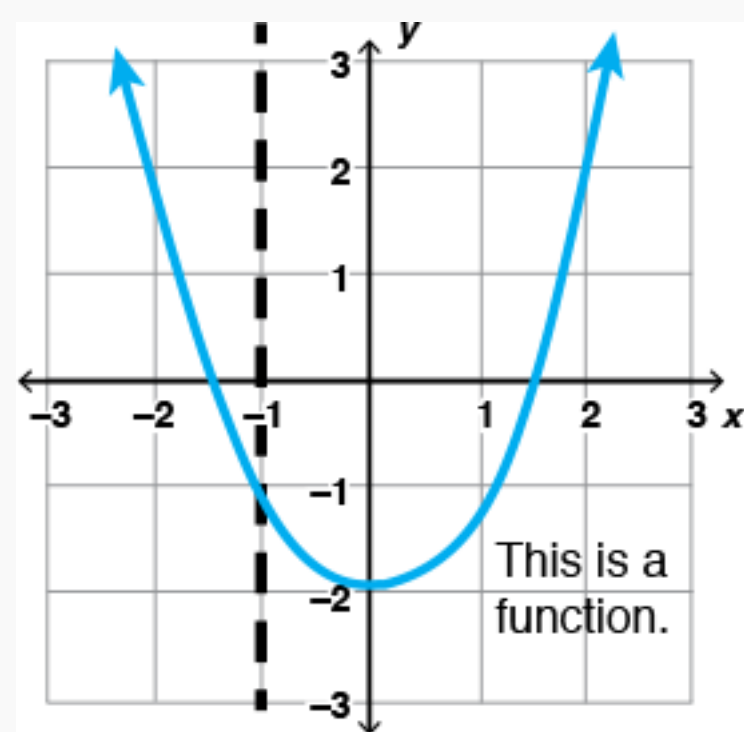
On each graph, the blue line represents a relation. Is the relation a function?

Imagine drawing a vertical line through each graph.

If the vertical line passes through the graph more than once, then there is an x -coordinate with more than one possible y -coordinate.



The vertical line crosses the blue line more than once.

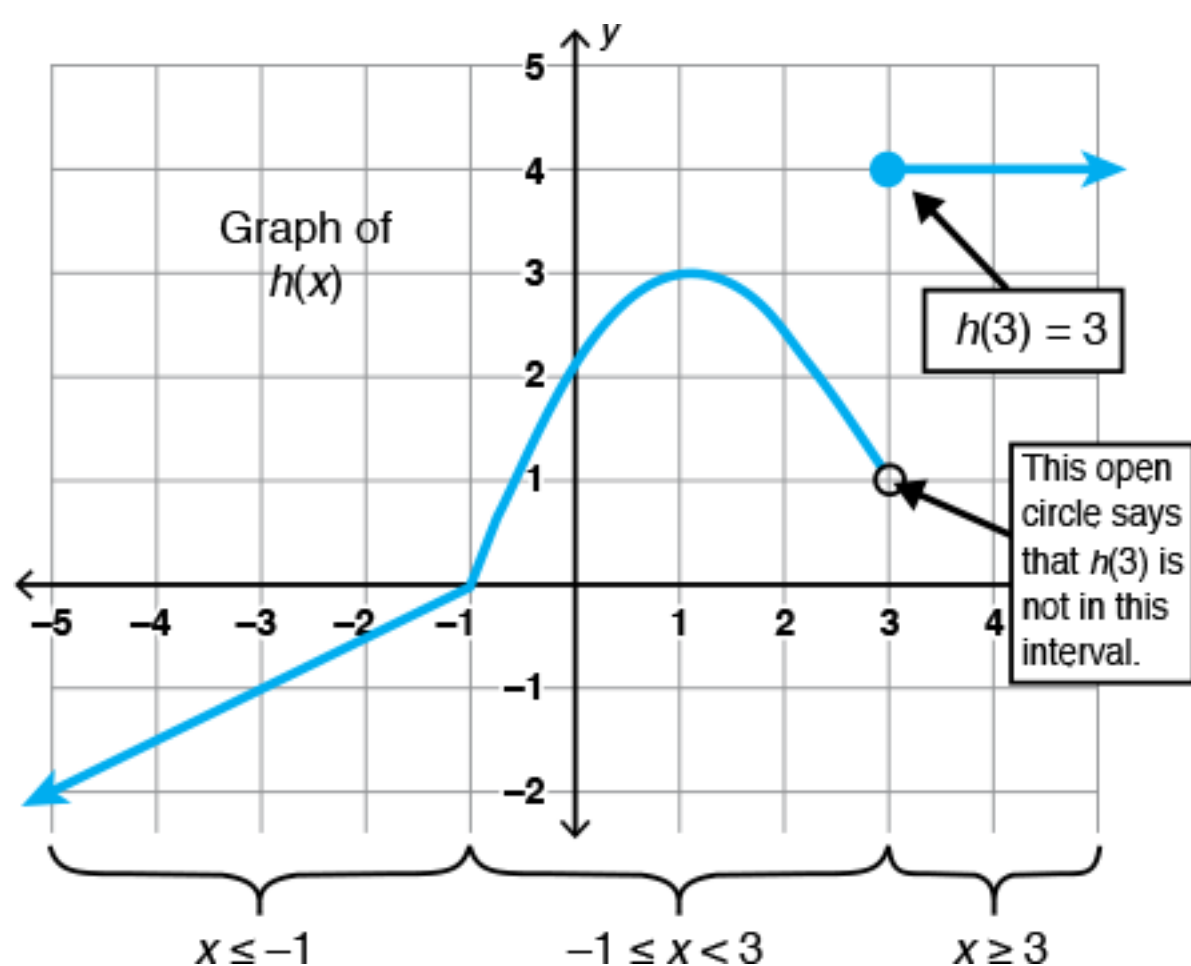


No matter where you place the vertical line, it will cross the blue line only once.

You may be asked a question about a specific interval on a graph. An **interval** is the distance between two points, usually on the x -axis.

The graph of function $h(x)$ has several parts. Notice how intervals can be used to describe and analyze the graph.

- During the interval $x \leq -1$, $h(x)$ increases as x increases.
- In the interval $-1 \leq x < 3$, the function reaches its highest point when $x = 1$.
- The function does not change in the interval $x \geq 3$.



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Lesson 5: Functions

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